

# Alberta opioid deaths around COVID-19

An interrupted time series with placebo and NB cross-checks

Adilbek Sultanov · personal learning project · May 2026

[github.com/Adilbek123/alberta-substance-use-trends](https://github.com/Adilbek123/alberta-substance-use-trends)

# What I asked, and how I tried to answer it

## Question

Did Alberta's opioid death rate shift at a level higher than the pre-COVID trend can explain, starting from 2020 Q2 (first full quarter under the public health emergency)?

## Data

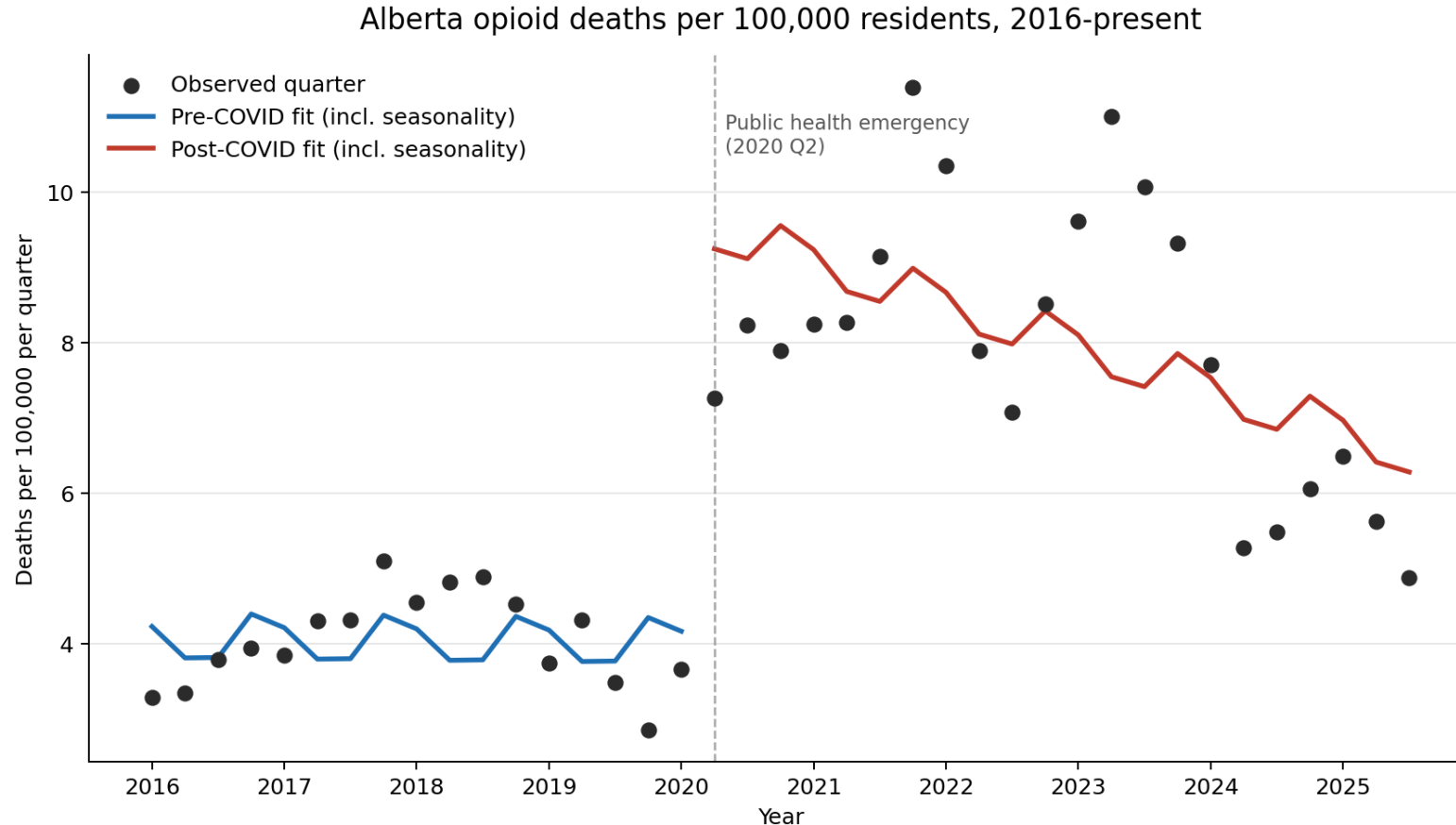
PHAC Health Infobase substance-related harms data + StatCan Table 17-10-0009-01. 39 quarters of Alberta opioid deaths, 2016 Q1 to 2025 Q3, normalised to deaths per 100k residents.

## Method

Interrupted time series with a level and slope change around 2020 Q2. Quarter fixed effects for seasonality. Newey-West HAC SE with small-sample correction.

Robustness: HAC lag sensitivity, donut (drop 2020 Q1), seven placebo cutoffs in the pre-period, negative binomial cross-check with log-population offset.

# Sharp, large, robust break at 2020 Q2



Sources: PHAC Health Infobase substance-related harms data; StatCan Table 17-10-0009-01.  
Level shift at 2020 Q2 (segmented regression with quarter FE, HAC SE 3-lag, small-sample correction): +5.50 per 100k [95% CI +3.60, +7.39].

## Main spec

Level shift at 2020 Q2:

**+5.50 per 100k per quarter**

(95% CI +3.60, +7.39,  $p < 0.001$ )

At mean AB pop: ~+246 deaths/quarter

## Robustness

HAC L=1..6: +5.50 (SE 0.79–0.98)

Donut (drop 2020 Q1): +5.34

NB rate ratio: 2.43 (1.80, 3.26)

## Placebo cutoffs

Seven fake cutoffs in pre-period:

All point opposite direction.

Largest |placebo|: 1.86

Real jump (+5.50) is ~3× larger.

# What this identifies — and what it doesn't

## What we can claim

A sharp, large, robust break in Alberta opioid mortality at 2020 Q2.

Pre-trend flat, placebos null, donut and NB cross-check consistent. This is a real shift in the data, not an artefact of specification choices.

## What we cannot claim

That COVID-19 itself caused the jump.

Many things moved at the same cutoff: the virus, public health emergency response, supply chain for the unregulated drug supply, harm reduction service capacity, mental health services, social isolation, and economic shock. ITS with a single cutoff cannot separate them.

## Next steps to sharpen identification

1. Difference-in-discontinuities using other provinces with different pandemic-response timing
2. Synthetic control to build a counterfactual Alberta without COVID
3. Stratification by age, sex, zone, and substance type to localise where the shift concentrated